



DevOps Open Agent

Open Source AI-Powered DevOps Troubleshooting Platform

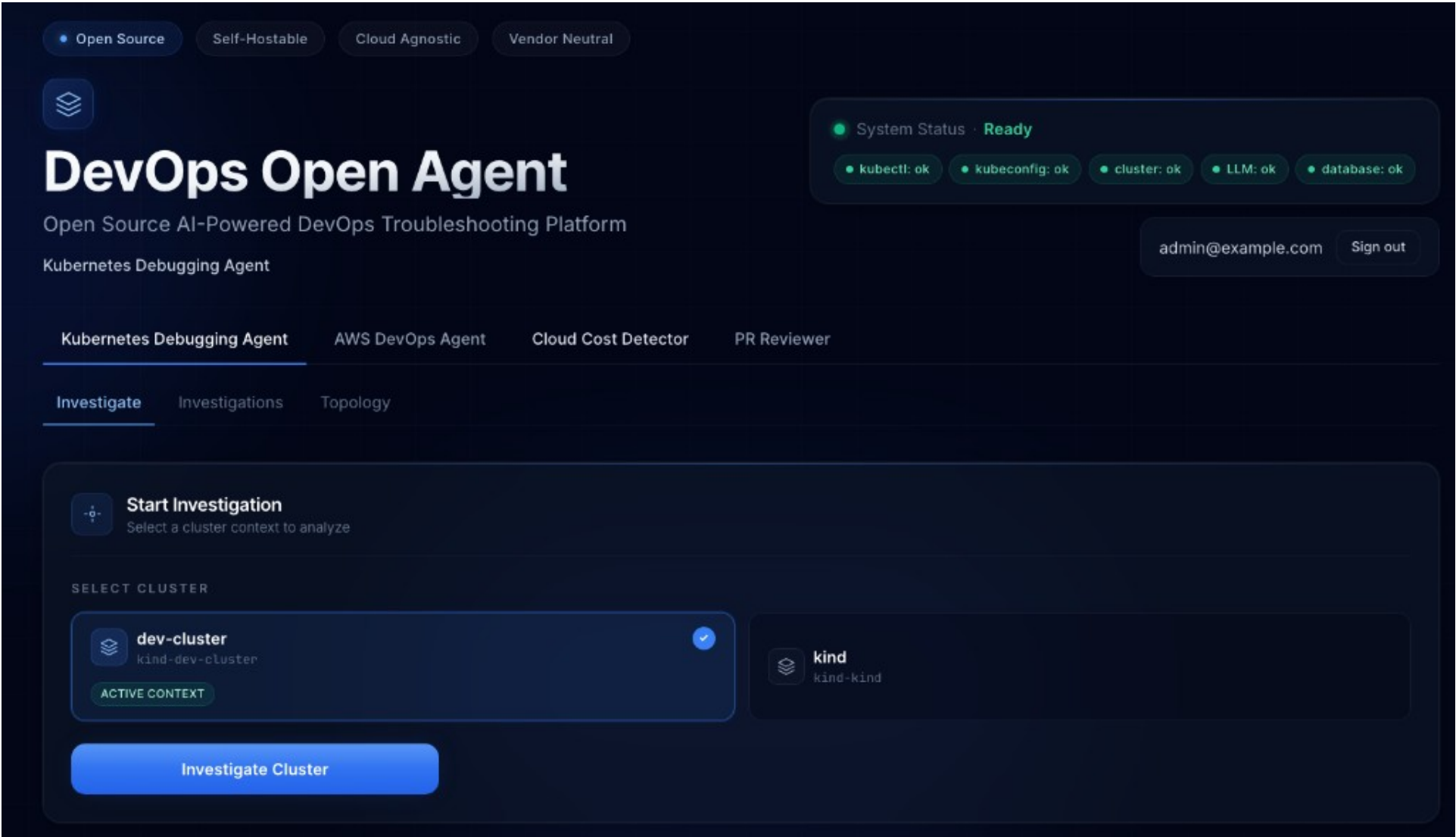
Self-hostable platform for Kubernetes debugging, AWS troubleshooting, cloud cost optimization, and AI-powered GitHub PR review. Run locally or on your own infrastructure with Docker Compose.

Modules

1. Kubernetes Debugging Agent — clusters, workloads, networking, topology
2. AWS DevOps Agent — EC2, VPC, security, load balancers, CloudTrail
3. Cloud Cost Detector — unused resources and savings estimates
4. PR Reviewer — DevOps-focused GitHub pull request reviews

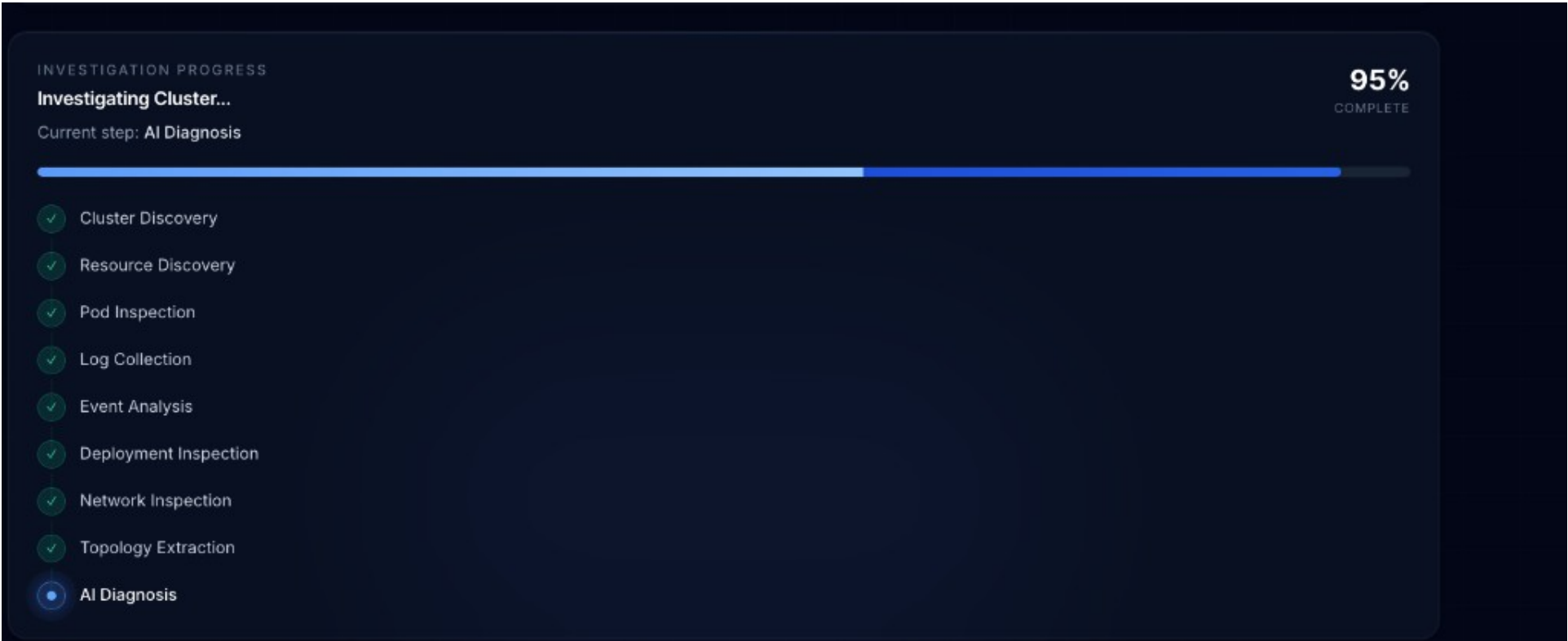
1. DevOps Open Agent

Platform home for the Kubernetes Debugging Agent. Select a cluster, check system readiness, and start an investigation.



2. Kubernetes Investigation

Live investigation progress across discovery, pods, logs, events, deployments, networking, topology, and AI diagnosis.



3. Kubernetes AI Diagnosis

AI root cause analysis with confidence score, evidence, and a clear summary of the ImagePullBackOff issue.

AI ANALYSIS

AI - Ollama

79%

CONFIDENCE

Diagnosis

ROOT CAUSE

The primary issue affecting the 'nginx' pod in the 'default' namespace is an incorrect image reference specified in its deployment/pod definition. The container attempts to pull the non-existent image tag 'docker.io/library/nginx1:latest'. This specific repository and tag combination does not exist or requires authorization that the cluster cannot provide, leading to repeated 'ImagePullBackOff' errors.

SUMMARY

The 'nginx' pod is stuck in a Pending state due to an ImagePullBackOff error. The root cause is a typo or incorrect image name ('nginx1' instead of 'nginx') used in the container specification. This prevents Kubernetes from pulling the necessary application image, causing repeated failures and back-off cycles.

EVIDENCE

PODS

The pod 'nginx' is in Pending status with reason ImagePullBackOff, indicating a failure to pull the required container image.

PER-POD ISSUES

The detailed message confirms the failure: 'failed to pull and unpack image "docker.io/library/nginx1:latest": failed to resolve reference... pull access denied, repository does not exist or may require authorization'.

EVENT FINDINGS

Multiple events confirm the failure pattern with 'BackOff' and 'ImagePullBackOff' warnings for the pod/nginx object.

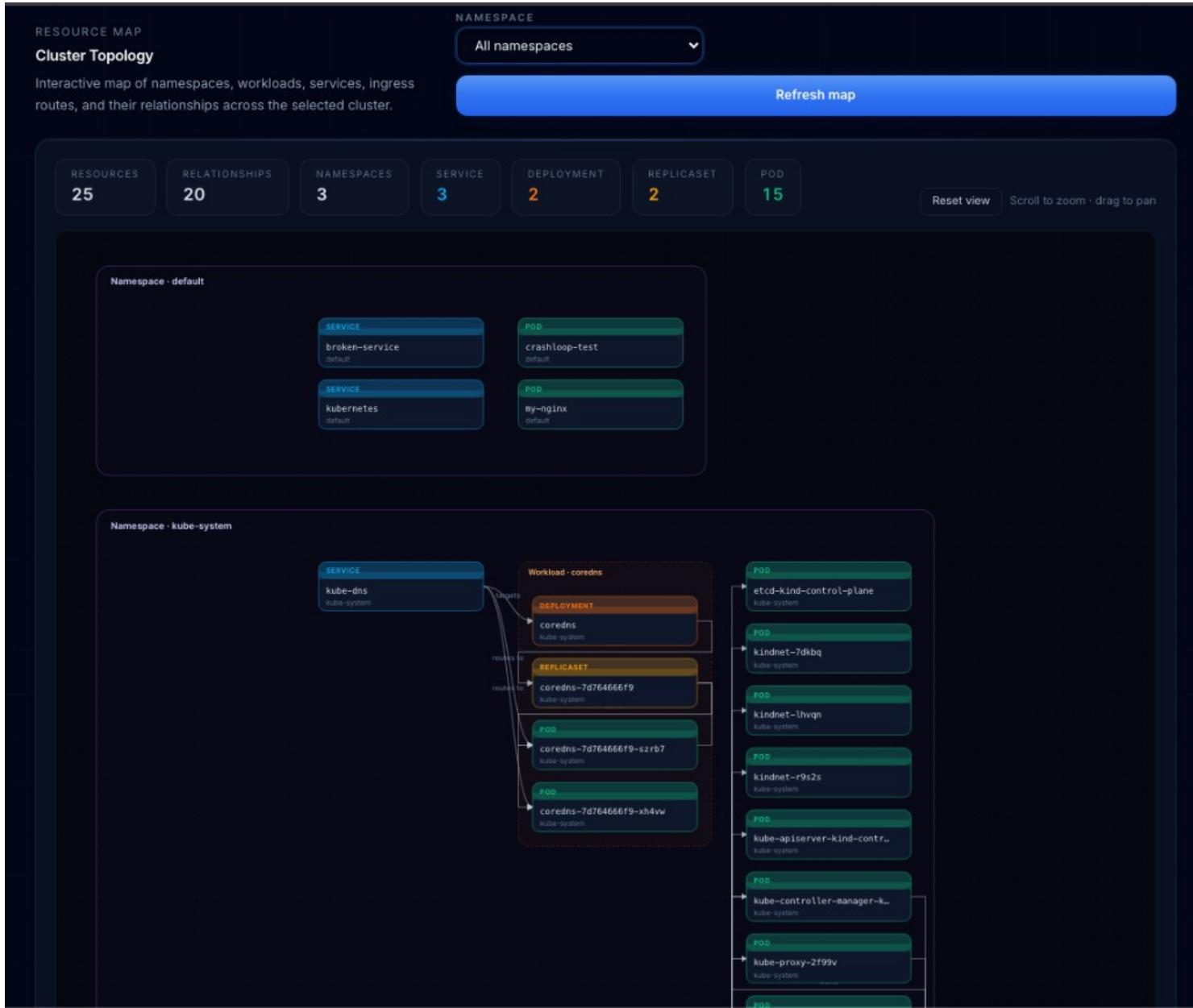
4. Recent Investigation History

Unified history showing root cause, agent, cluster, status, confidence, and timestamps for past runs.

HISTORY						
Recent Investigations						24 total
ROOT CAUSE	AGENT	CLUSTER	STATUS	CONFIDENCE	STARTED	
The primary issue affecting the 'nginx' pod is an incorrect image reference specified in its...	Kubernetes	kind-dev-cluster	success	79%	6/23/2026, 9:00:52 AM	View →
The primary issue affecting the 'nginx' pod in the 'default' namespace is an incorrect imag...	Kubernetes	kind-dev-cluster	success	79%	6/23/2026, 8:59:13 AM	View →
1. The 'nginx' pod in the 'default' namespace is in a 'Pending' state due to an 'ErrImagePull'...	Kubernetes	kind-dev-cluster	success	90%	6/21/2026, 8:10:31 PM	View →
1. The nginx pod is in a Pending state due to an ErrImagePull error, which indicates that the...	Kubernetes	kind-dev-cluster	success	85%	6/21/2026, 8:04:36 PM	View →
The primary issue affecting the 'create-container-error' pod is a misconfiguration in...	Kubernetes	kind-dev-cluster	success	79%	6/21/2026, 7:06:38 PM	View →
The primary issue affecting the 'config-error-pod' in the 'default' namespace is a missing...	Kubernetes	kind-dev-cluster	success	95%	6/21/2026, 7:01:25 PM	View →
The primary issue preventing the 'pending-pod' from running is resource starvation on t...	Kubernetes	kind-dev-cluster	success	95%	6/21/2026, 6:50:30 PM	View →
2 independent pod issues were detected ('crashloop-test', 'my-nginx'). Each issue has...	Kubernetes	kind-kind	success	79%	6/21/2026, 6:40:06 PM	View →

5. Kubernetes Cluster Topology

Namespace-grouped resource map of services, deployments, replica sets, and pods with relationship edges.



6. AWS DevOps Agent

Choose AWS account, region, and troubleshooting category such as full scan, security, EC2, network, or change audit.

Kubernetes Debugging AgentAWS DevOps AgentCloud Cost DetectorPR Reviewer

InvestigateInvestigationsTopology

**Troubleshoot AWS Infrastructure**
Choose what to investigate, then collect evidence and run AI analysis

SELECT AWS ACCOUNT

696364089559696364089559

SELECT REGION

ap-northeast-1ap-northeast-2ap-northeast-3ap-south-1ap-southeast-1ap-southeast-2ca-central-1eu-central-1eu-north-1eu-west-1eu-west-2eu-west-3sa-east-1us-east-1us-east-2us-west-1us-west-2

WHAT DO YOU WANT TO TROUBLESHOOT?

Full Scan
Detect all issues — security, EC2, network, load balancers, alarms, and changes

Security & Exposure
Open security groups, internet exposure, and who changed rules (CloudTrail)

EC2 Availability
Stopped instances, health checks, and who stopped them

Network & Connectivity
VPC, subnets, routing, and connectivity-blocking rules

Load Balancers
Unhealthy targets, inactive load balancers, routing issues

Performance & Alarms
CloudWatch alarms, metric anomalies, and capacity signals

Change Audit
Who changed what and when — CloudTrail and Config timeline

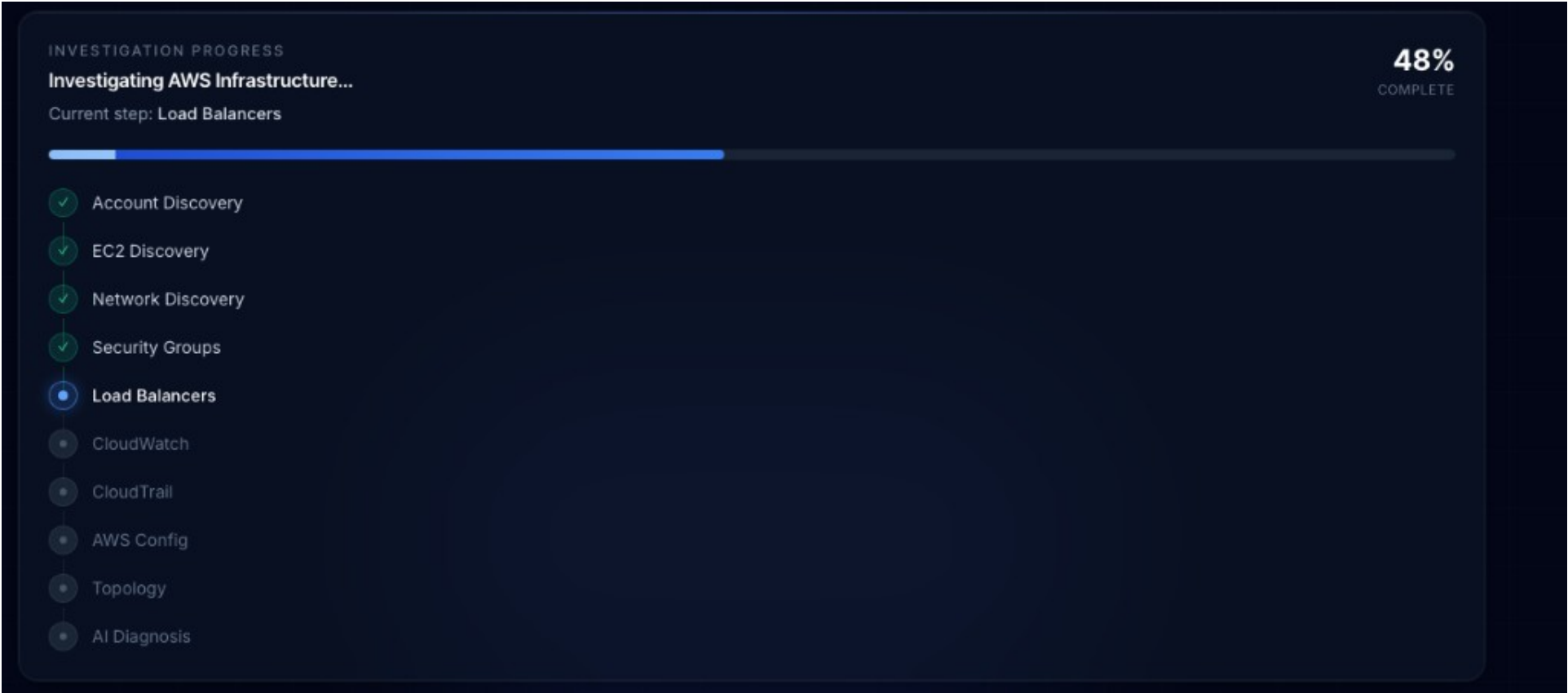
DESCRIBE THE ISSUE (OPTIONAL)

e.g. I opened HTTP to the internet on a security group and want to know if it's exposed...

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7. AWS Investigation

AWS investigation pipeline covering EC2, network, security groups, load balancers, CloudWatch, CloudTrail, Config, topology, and AI diagnosis.



8. AWS Investigation History

AWS investigation history with account/region, status, confidence, and root cause summaries.

• Open Source

Self-Hostable

Cloud Agnostic

Vendor Neutral



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AWS DevOps Agent

System Status · Ready

kubectl: ok

kubeconfig: ok

cluster: ok

LLM: ok

database: ok

admin@example.com

Sign out

Kubernetes Debugging Agent

AWS DevOps Agent

Cloud Cost Detector

PR Reviewer

Investigate

Investigations

Topology

HISTORY

Recent Investigations

2 total

ROOT CAUSE	AGENT	ACCOUNT / REGION	STATUS	CONFIDENCE	STARTED	
Investigation in progress / no RCA	AWS	6963640889559/ap-northeast-1	running	—	6/23/2026, 9:04:18 AM	View →
Multiple high-severity issues detected, including internet-exposed security group...	AWS	6963640889559/us-west-2	success	85%	6/22/2026, 9:18:40 AM	View →

9. AWS AI Analysis

AI diagnosis for exposed security groups, stopped EC2 instances, and actionable suggested fixes with CLI examples.

AI ANALYSIS

AI - OpenAI

85%
CONFIDENCE

Diagnosis

ROOT CAUSE

Multiple high-severity issues detected, including internet-exposed security group rules and a stopped EC2 instance.

SUMMARY

The EC2 instance 'ideaweaver-web-server' (i-0910dcb7d868d789b) is stopped, and security group 'launch-wizard-2' (sg-00cfd6e0a1ce89f31) has ingress rules allowing traffic on ports 80 and 443 from 0.0.0.0/0, exposing it to the internet. Additionally, CloudTrail and CloudWatch evidence is limited to the last 24 hours, which may omit older changes.

EVIDENCE

EC2

Instance i-0910dcb7d868d789b is stopped and was stopped by user plakhera on 2026-06-22T04:04:16+00:00.

SECURITY_GROUPS

Security group sg-00cfd6e0a1ce89f31 allows ingress on tcp port 80 from 0.0.0.0/0.

SECURITY_GROUPS

Security group sg-00cfd6e0a1ce89f31 allows ingress on tcp port 443 from 0.0.0.0/0.

SECURITY_GROUPS

Security group sg-0976fe75497cd5777 allows ingress on tcp port 443 from 0.0.0.0/0.

CLOUDTRAIL

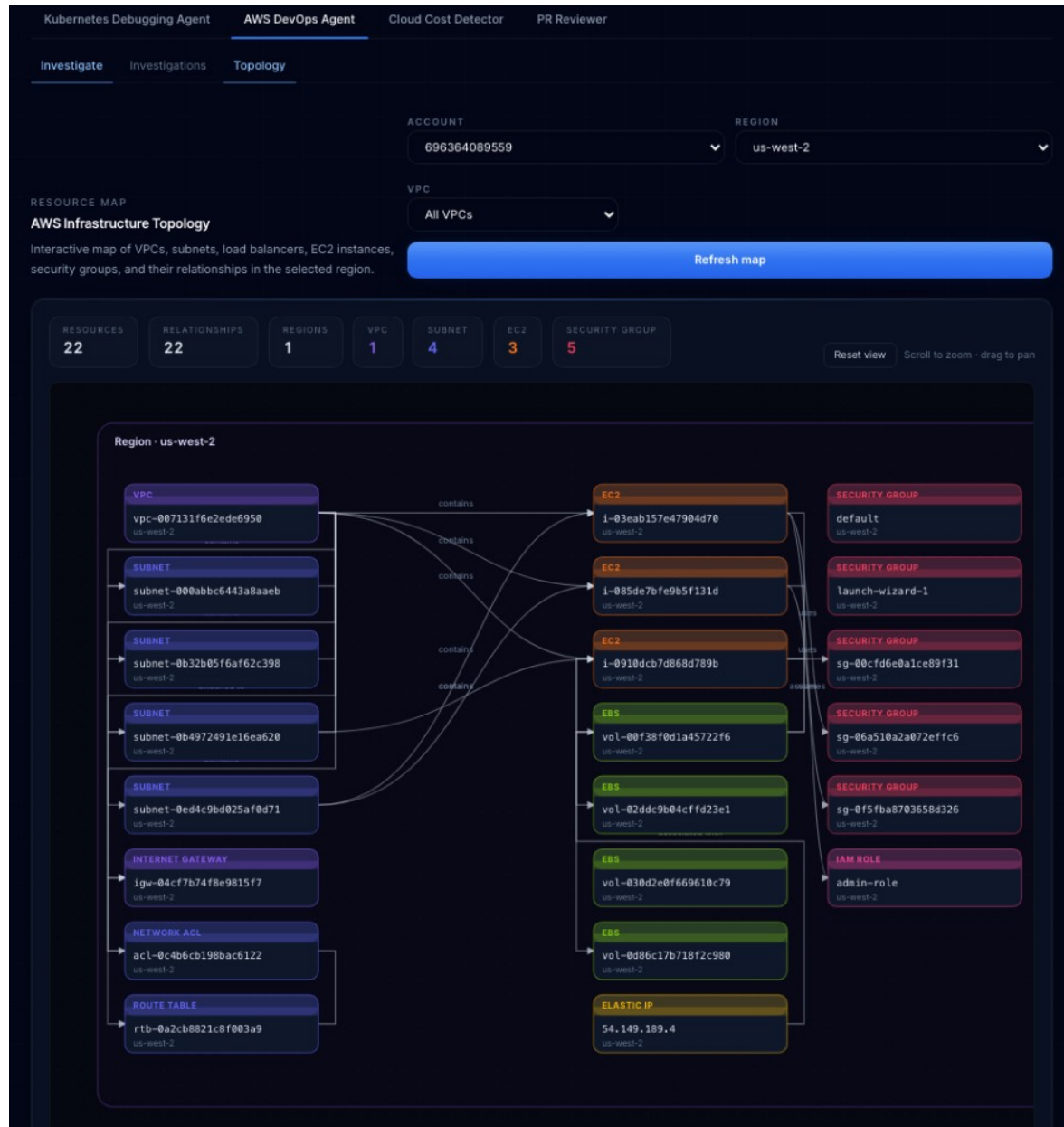
CloudTrail evidence is limited to the last 24 hours.

SUGGESTED FIX

1. ****Stop Internet Exposure****: Modify security group sg-00cfd6e0a1ce89f31 to restrict ingress on ports 80 and 443. Remove the rules allowing access from 0.0.0.0/0.
- Command: `'aws ec2 revoke-security-group-ingress --group-id sg-00cfd6e0a1ce89f31 --protocol tcp --port 80 --cidr 0.0.0.0/0'`
- Command: `'aws ec2 revoke-security-group-ingress --group-id sg-00cfd6e0a1ce89f31 --protocol tcp --port 443 --cidr 0.0.0.0/0'`

10. AWS Topology

Interactive AWS topology map for VPCs, subnets, EC2, EBS, security groups, gateways, and IAM relationships.



11. Cloud Cost Detector

Multi-step AWS cost optimization workflow from discovery through unused resource analysis to AI cost analysis.

DevOps Open Agent

Open Source AI-Powered DevOps Troubleshooting Platform

Cloud Cost Detector

System Status: Ready

kubectx: okkubernetes: okcluster: okLLM: okdatabase: ok

admin@example.comSign out

Kubernetes Debugging AgentAWS DevOps AgentCloud Cost DetectorPR Reviewer

AnalyzeInvestigations

\$Cloud Cost Detector

AWS Cost Optimization Platform

ACCOUNT

696364089559

REGION

us-west-2

Analyzing...

INVESTIGATION PROGRESS

Analyzing AWS Cost Optimization Opportunities...

Current step: AI Cost Analysis

95% COMPLETE

✓ Account Discovery

✓ Resource Discovery

✓ Unused Resource Analysis

✓ Cost Estimation

● AI Cost Analysis

12. Cloud Investigation Details

Detailed savings estimate, Cost Explorer context, AI optimization report, and prioritized findings for wasteful resources.

Investigation Details

ab6ce0d5-be2a-428c-80c5-d708c6e15b10

AGENT	ACCOUNT / REGION	STATUS	AI PROVIDER	CONFIDENCE	STARTED
Cloud Cost	696364089559/us-west-2	Success	Ollama	88%	6/22/2026, 10:55:29 AM

COST ESTIMATION

\$3.20 potential monthly savings

Estimated \$3.20/month recoverable from 2 unused or wasteful resource(s) using AWS list pricing. Regional spend (last 30 days): \$3.60 via Cost Explorer.

PER-RESOURCE ESTIMATES

i-0910dcb7d868d789b

\$2.40/mo · medium confidence

Stopped instance compute is \$0; estimate reflects attached EBS storage that could be eliminated if the instance is terminated.

vol-030d2e0f669610c79

\$0.80/mo · high confidence

10 GB gp3 at \$0.080/GB-month.

REGIONAL SPEND (COST EXPLORER)

\$3.60

2026-05-23 → 2026-06-22

Amazon Virtual Private Cloud

\$3.60

EC2 - Other

—

AmazonCloudWatch

—

Amazon Elastic Compute Cloud - Compute

—

Amazon Simple Storage Service

—

AI COST ANALYSIS

Optimization Report

AI · Ollama

SUMMARY

The analysis identified two primary areas for cost optimization: a stopped EC2 instance and an unattached EBS volume. The stopped EC2 instance, i-0910dcb7d868d789b, is incurring charges for attached storage/IPs even when compute is off. The unattached EBS volume, vol-030d2e0f669610c79, is a clear candidate for deletion as it provides no apparent value. Immediate action should focus on validating the need for these resources.

OVERALL RISK

medium

ESTIMATED MONTHLY SAVINGS

\$3

Confidence: high · Based on 2 rule-based finding(s) (2 unused, 0 underutilized) including 1 high-severity item(s) from live AWS discovery in the selected region. The analysis references 3 discovered resource ID(s). Confidence increased because the analysis cites 3 discovered resource(s). Estimated monthly savings: \$3.20 (high confidence) from AWS list pricing and Cost Explorer context.

FINDINGS (2)

Unattached EBS Volume Candidate for Deletion

vol-030d2e0f669610c79

EBS · confirmed

high

confirmed

validate first

The volume vol-030d2e0f669610c79 is explicitly listed as 'available' and unattached (attached_instance_id: null). It is incurring charges for 10 GB of provisioned storage. If this volume contains no necessary data or backups, it should be deleted immediately to stop billing for the provisioned storage.

Est. savings: \$1/month · high confidence

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13. GitHub PR Reviewer

Configure GitHub webhooks and tokens, or trigger a manual DevOps PR review for any repository.



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PR Reviewer

System Status · Ready

kubectll: okkubernetesconfig: okcluster: okLLM: okdatabase: ok

admin@example.comSign out

Kubernetes Debugging AgentAWS DevOps AgentCloud Cost DetectorPR Reviewer

ReviewInvestigations

PR Reviewer

AI-powered DevOps review for GitHub Pull Requests

Setup Instructions

Configure a GitHub webhook on your repository to trigger automatic DevOps PR reviews.

Webhook URL

http://localhost:8080/api/v1/pr-reviewer/webhook

Content type

application/json

Events

pull_request (opened, synchronize, reopened)

Secret

Set GITHUB_WEBHOOK_SECRET in backend .env and use the same value in GitHub.

GitHub token

Set GITHUB_TOKEN with repo scope to fetch PR files and post review comments.

Manual Review

Trigger a DevOps review without waiting for a webhook event.

Owner

ideaweaver-ai

Repository

devops-testing

Pull Request Number

1

Review Pull Request

14. PR Review AI Analysis

Completed AI DevOps PR review with risk rating and structured security and reliability findings.

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ReviewInvestigations

← Back to Investigations

Review Detail

Pull Requestideaweaver-ai/pr-review-repo #1Create logfile.pyView GitHub comment

Status: CompletedRisk: HighFindings: 3

Review

DevOps PR Review

Summary

The new `logfile.py` script is intended for cleaning up log directories but contains several critical reliability and security flaws, parti

Overall Risk

High

Findings

1. [High] Security

File: `logfile.py`

Issue:

Unsafe default path for log directory.

Why it matters:

Using a hardcoded default like `/var/log` without proper validation or context can lead to unintended deletion of system files if the envir

Recommendation:

The script should validate that the provided `LOG\_DIR` (or its default) is within an expected, isolated application directory structure, or

Example fix:

Add checks to ensure the resolved log directory exists and is owned by the application process before proceeding with deletion.

2. [High] Reliability

File: `logfile.py`

Issue:

Unsafe recursive delete without safeguards.

Why it matters:

Using `shutil.rmtree(target, ignore\_errors=True)` is highly dangerous. If the path resolution fails or if the target directory contains cri

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